

PADALA RITHIKA LAKSHMI

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Software engineering student (Expected Graduation: May 2028) skilled in distributed systems, backend engineering, and cloud-native development. Experienced in building scalable, production-ready software through the full SDLC using Java, Python, and JavaScript in Agile/Scrum environments.

EDUCATION

DVR & Dr. HS MIC College of Technology

Kanchikacherla, Andhra Pradesh

B.Tech in Information Technology | CGPA: 9.50 / 10.00 | Expected Graduation: May 2028

Aug 2024 – May 2028

Coursework: Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, Computer Networks, DBMS (SQL, Relational Databases), Software Engineering, Complexity Analysis

TECHNICAL SKILLS

Languages: Java, Python, C++, JavaScript, SQL, HTML, CSS

Backend & Frameworks: FastAPI, Flask, Node.js, React, REST APIs, gRPC, Asynchronous Programming

Cloud & DevOps: Google Cloud Platform (GCP), Docker, CI/CD, Linux, Git, Version Control

Databases: PostgreSQL, MongoDB, Redis, SQL, Relational Database Design

Testing: Unit Testing, Integration Testing, API Testing, Test Automation, Debugging, Error Handling

Core CS: Distributed Systems, Object-Oriented Design, Software Architecture, Concurrent Programming, Multi-Threading, Scalability, Performance Optimization, Agile/Scrum, SDLC

SOFTWARE ENGINEERING PROJECTS

Distributed Key-Value Store with Raft Consensus

Nov 2025 – Feb 2026

Python, gRPC, Multi-Threading, Unit Testing

- Engineered a fault-tolerant distributed key-value store across 5 nodes by implementing Raft consensus for leader election, log replication, and consistent state management
- Optimized concurrent request handling using Python threading with synchronization primitives (Locks, RLocks, Conditions), achieving 3,000+ ops/sec under 50-client concurrent load
- Automated reliability validation by building a chaos testing framework simulating network partitions and node failures; implemented write-ahead logging for crash recovery

Concurrent Task Scheduler with Work-Stealing

Feb 2026 – Mar 2026

Java, ExecutorService, Thread Pools, JUnit, Object-Oriented Design

- Developed a concurrent task scheduler in Java with DAG-based dependency resolution, priority queues, and work-stealing across a configurable thread pool
- Implemented lock-free data structures using java.util.concurrent (ConcurrentHashMap, AtomicReference, CompletableFuture), achieving 3.8x speedup versus single-threaded baseline
- Collaborated with peers through Git-based code reviews; built real-time monitoring with structured logging and execution visualization using JavaScript

Scalable Code Search Engine – Full-Stack Cloud Application

Jan 2026 – Mar 2026

Python, FastAPI, React, JavaScript, HTML, CSS, PostgreSQL, MongoDB, Docker, GCP

- Architected and deployed a full-stack code search application on Google Cloud Run with FastAPI backend, React frontend, and Docker-based CI/CD deployment
- Scaled ingestion pipeline to process 10,000+ files using multi-threading and Redis caching, achieving sub-150ms API latency under 100 concurrent users
- Implemented integration tests for REST API endpoints and configured production monitoring; collaborated in a 3-member Agile team with sprint-based delivery

OPEN SOURCE EXPERIENCE

JSON Schema (json-schema-org)

2026

Open Source Contributor

- Studied and navigated the JSON Schema specification codebase to understand modular architecture, validation logic, and cross-language design patterns
- Explored Git-based open source collaboration workflows including issue tracking, branching strategies, and pull request review processes

CERTIFICATIONS

- Google Cloud Certified – Cloud Computing Foundations and Infrastructure (Google Cloud Skills Boost)
- Machine Learning Specialization – Andrew Ng, Stanford University (Coursera)

ACHIEVEMENTS

- Flipkart Grid 8.0** – Advanced to the selection round among 100,000+ nationwide participants (2026)
- Amazon ML Summer School 2026** – Qualified for selection test from a nationwide applicant pool
- Infosys Springboard** – Selected for Virtual Internship 7.0 through nationwide competitive evaluation
- Hackathon Winner** – Won 3 inter-college hackathons as team lead: AI for Bharat, GenAI Exchange, Sitestrom
- Academic Excellence** – CGPA 9.50/10; consistently ranked in top 5% of department (200+ students)